iLearn Practice Exam

10BR · Chapter / Topic 6

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Chapter 6: Automated and Emerging Technologies

Name: _____

Class: _____

Date: _____

Time allowed: 40 minutes

TOTAL MARKS: 33

Marks obtained: ..... / 33

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 33.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 4 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.
· Only 33 marks of past-paper questions are available for this chapter, so the recommended time is 40 minutes (1 min 1

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 A garage uses an expert system to help diagnose any problems with cars that need repair.

(a) The expert system is an example of artificial intelligence (AI).

Describe what is meant by AI.

.....

.....

.....

..... [2]

(b) A car has a problem with its braking system, so it is brought into the garage.

Explain how the expert system operates and how it is used to help diagnose the problem.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

2 A hospital has computers and technology that it uses when caring for its patients.

(a) One computer is an expert system that is used to help diagnose a patient's illness.

The doctor inputs the patient's symptoms into an interface and a diagnosis is output.

Describe how the expert system decides the diagnosis.

.....

.....

.....

.....

.....

..... [3]

(b) The hospital has a robot that is used to perform surgery. A doctor controls the robot from a different location to perform surgery on a patient.

(i) One feature of the robot is that it has electrical components.

Give **two** examples of the electrical components of a robot.

1 .....

2 .....

[2]

(ii) Explain the advantages of using the robot to perform surgery.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(iii) Explain **one** disadvantage of using the robot to perform surgery.

.....

.....

.....

..... [2]

3 A child has a robot animal. The robot animal can perform actions like making a noise and walking around a room.

If the robot animal hears a clapping sound, it makes a noise.

If the robot animal is walking and it senses an object is less than 10 cm away, it will stop.

(a) Give **one** suitable sensor that the robot animal could use to identify a clapping sound.

..... [1]

(b) Give **two** suitable sensors that the robot animal could use to identify if an object is nearby.

1 .....

2 .....
[2]

(c) One characteristic of the robot animal that makes it a robot is that it has electrical components that include sensors, a microprocessor and actuators.

(i) Give **one** other characteristic of the robot animal that makes it a robot.

..... [1]

(ii) The sensor continuously sends digitised data to the microprocessor.

Describe how the microprocessor uses the data from the sensor to stop the robot animal when an object is less than 10 cm away.

.....

.....

.....

.....

.....

..... [3]

(d) The robot animal uses artificial intelligence (AI).

Explain how the robot animal can use AI to learn where objects are in a room.

.....

.....

.....

.....

.....

.....

.....

..... [4]

4 A student has an application (app) on their smartphone that can be used to identify the name of a flower.

The student opens the app and takes a photograph of a flower. The photograph is stored as an image.

The app searches its database for similar images to find the name of the flower.

(a) The app uses artificial intelligence (AI) to identify the name of the flower.

Explain how the app can use AI to improve its ability to identify the correct name of the flower.

.....

.....

.....

.....

.....

.....

.....

..... [4]

END OF QUESTION PAPER